

Nonexistence of invertible cross-product twists for $\mathfrak{gl}_2(F)/(FI)$ in characteristic two

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Abstract

For every field F of characteristic two, the Lie algebra $L = \mathfrak{gl}_2(F)/(FI)$ cannot be represented on F^3 by the standard coordinate cross product followed by a fixed invertible linear map. The obstruction is the nonzero trace functional on L , which annihilates every commutator. A second proof compares the rank-two quotient bracket with the rank-three cross-product map on the exterior square. We also give an explicit singular rank-two twist over every such field, showing why invertibility is essential. The result applies to finite and infinite fields alike. The external attribution of the motivating question and the novelty of this elementary classification remain unverified.

1 The precise representation question

Throughout, F is a commutative field of characteristic two and I is the 2×2 identity matrix. We write

$$L = \mathfrak{gl}_2(F)/(FI), \quad [A + FI, B + FI] = AB - BA + FI.$$

Here $\mathfrak{gl}_2(F)$ is the vector space of all 2×2 matrices, equipped with the commutator bracket. Scalar matrices are central, so this quotient is a well-defined three-dimensional Lie algebra. In particular, the quotient is taken in $\mathfrak{gl}_2(F)$, not in $\mathfrak{sl}_2(F)$.

For column vectors $u, v \in F^3$, the standard coordinate cross product is

$$u \times v = \begin{pmatrix} u_2v_3 - u_3v_2 \\ u_3v_1 - u_1v_3 \\ u_1v_2 - u_2v_1 \end{pmatrix}. \quad (1)$$

This bilinear map is alternating, also in characteristic two, where minus and plus coincide. The question considered here asks whether there exist an F -linear isomorphism $\phi : L \rightarrow F^3$ and a matrix $E \in \mathrm{GL}_3(F)$ such that

$$\phi([x, y]) = E(\phi(x) \times \phi(y)) \quad \text{for all } x, y \in L. \quad (2)$$

Both ϕ and E may depend on F , but one fixed E must work for every pair x, y .

The supplied question attributes this formulation to Question 3.7 of arXiv:2609.05386v1 [1], and reports a construction in characteristic different from two. The cited text has not been inspected; neither that comparison nor agreement with the invertibility requirement in (2) has been verified. We therefore resolve only the explicit formulation above and make no claim of priority, novelty, or resolution of the externally attributed question. As broad adjacent context for exact matrix-rank and field-dependent representation questions, we cite Zeng [2]. That reference is not used as a result about the Lie algebra or cross-product model studied here.

2 The trace obstruction

Theorem 1. *For every field F of characteristic two, there are no F -linear isomorphism $\phi : \mathfrak{gl}_2(F)/(FI) \rightarrow F^3$ and $E \in \mathrm{GL}_3(F)$ satisfying (2). Thus the class of characteristic-two fields admitting the required invertible-twist representation is empty.*

Proof. The formula

$$\tau : L \longrightarrow F, \quad \tau(A + FI) = \mathrm{tr}(A)$$

defines a linear functional: changing A to $A + tI$ changes its trace by $2t = 0$. It is nonzero because $\tau(E_{11} + FI) = 1$, where E_{ij} denotes the usual matrix unit. Commutativity of F gives

$$\mathrm{tr}(AB) = \sum_{i,j} A_{ij}B_{ji} = \sum_{i,j} B_{ij}A_{ji} = \mathrm{tr}(BA).$$

Consequently $\tau([x, y]) = 0$ for every $x, y \in L$.

Suppose that ϕ and E satisfy (2). Then $\lambda = \tau \circ \phi^{-1}$ is a nonzero linear functional on F^3 . Since ϕ is onto, the representation identity implies

$$\lambda(E(u \times v)) = 0 \quad \text{for all } u, v \in F^3.$$

For the standard basis e_1, e_2, e_3 of F^3 , formula (1) gives

$$e_2 \times e_3 = e_1, \quad e_3 \times e_1 = e_2, \quad e_1 \times e_2 = e_3.$$

Hence λE vanishes on a basis and is zero. Invertibility of E then forces $\lambda = 0$, a contradiction. $\square$

In particular, $F = \mathbb{F}_2$ already refutes the assertion that every characteristic-two field admits such a representation. The proof also excludes every other such field; it uses no finiteness, perfection, or algebraic-closure assumption.

3 Bracket rank and the singular boundary

The quotient bracket gives another proof and an explicit description of what changes when invertibility is dropped. Define

$$q : M_2(F) \longrightarrow F^3, \quad q \begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a + d, b, c)^\top.$$

The map is onto, and its kernel is exactly FI , since $a + d = 0$ means $a = d$ in characteristic two. Let $\bar{q} : L \rightarrow F^3$ be the induced isomorphism. The classes

$$h = E_{11} + FI, \quad e = E_{12} + FI, \quad f = E_{21} + FI$$

form a basis, and direct multiplication yields

$$[h, e] = e, \quad [h, f] = f, \quad [e, f] = 0. \tag{3}$$

For the last relation, $[E_{12}, E_{21}] = E_{11} - E_{22} = I$ in characteristic two, so its class in L is zero. Thus $[L, L] = \mathrm{span}_F\{e, f\} = \ker \tau$ has dimension two.

A second proof of Theorem 1. Because the bracket and cross product are alternating, they induce linear maps $B : \Lambda^2 L \rightarrow L$ and $C : \Lambda^2 F^3 \rightarrow F^3$. With domain bases $(h \wedge e, h \wedge f, e \wedge f)$ and $(e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3)$, respectively, their matrices are

$$B = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad C = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

They have ranks two and three. Identity (2) would give

$$\phi \circ B = E \circ C \circ (\Lambda^2 \phi). \quad (4)$$

The exterior-square map $\Lambda^2 \phi$ is invertible, with inverse $\Lambda^2(\phi^{-1})$. The two sides of (4) therefore have ranks two and three when E is invertible, which is impossible. This argument uses alternation and does not divide by two.

Proposition 2. *Over every field F of characteristic two, the isomorphism $\bar{q}$ gives a representation of the form (2) with the singular matrix*

$$E_0 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Moreover, any matrix $E \in M_3(F)$ satisfying that identity with an F -linear isomorphism ϕ necessarily has rank two.

Proof. In the coordinates $(r, s, t)^\top = \bar{q}(x)$ and $(r', s', t')^\top = \bar{q}(y)$, relations (3) give

$$\bar{q}([x, y]) = \begin{pmatrix} 0 \\ rs' + sr' \\ rt' + tr' \end{pmatrix} = E_0(\bar{q}(x) \times \bar{q}(y)).$$

The last equality follows directly from (1) in characteristic two. The two nonzero columns of E_0 are independent. For any other such representation, equation (4) and invertibility of ϕ , C , and $\Lambda^2 \phi$ force $\text{rank}(E) = \text{rank}(B) = 2$. $\square$

The singular representation does not satisfy the invertibility hypothesis of Theorem 1. It explains why that hypothesis must be retained when comparing different meanings of a twisted cross product.

4 Exact checks and assistance disclosure

The accompanying verification materials supplement the algebraic proofs. One Python program checks generic matrix identities in $\mathbb{F}_2[a, b, c, d, e, f, g, h, s, t]$: trace vanishing on commutators, invariance under scalar changes of representatives, and the displayed singular-twist identity. Polynomial monomials retain their multiplicities; these are polynomial identities, not just evaluations on Boolean inputs.

The finite checks cover exactly $\mathbb{F}_2$. They enumerate all 256 ordered pairs of 2×2 matrices and all $|\text{GL}_3(\mathbb{F}_2)| = (8-1)(8-2)(8-4) = 168$ coordinate changes. For each coordinate change, the three cyclic basis products determine the only possible twist. Each resulting matrix has rank two, and its identity is checked on all 64 ordered vector pairs, for 10,752 transformed pair checks in total. A separately written verifier reads the serialized matrix certificate and repeats these finite checks using integer arithmetic modulo two and bit-vector

span enumeration, without importing the first program. The accompanying reproducibility guide gives the commands and the recorded coverage. No finite computation is used to infer the arbitrary-field theorem.

AI assistance was used for proof development, exact-verification code, mathematical review, and manuscript preparation. A separate AI review pass reconstructed both proofs and independently implemented the finite verifier. This describes separate implementations and review contexts; it is not a claim of external peer review or statistical independence. The mathematical arguments above are self-contained and require no result from the cited literature. External attribution and broader prior-work comparison remain unverified as stated in the opening section.

References

- [1] arXiv:2609.05386v1. <https://arxiv.org/abs/2609.05386v1>. The supplied question attributes the motivating problem to Question 3.7; the source text and its bibliographic metadata have not been independently verified.
- [2] Zijian Zeng, *A Rational Rank-(2, 2) Hadamard Factorization of a Proposed 4×4 Real Counterexample*, preprint, Zenodo, September 2026. <https://doi.org/10.5281/zenodo.22242517>.